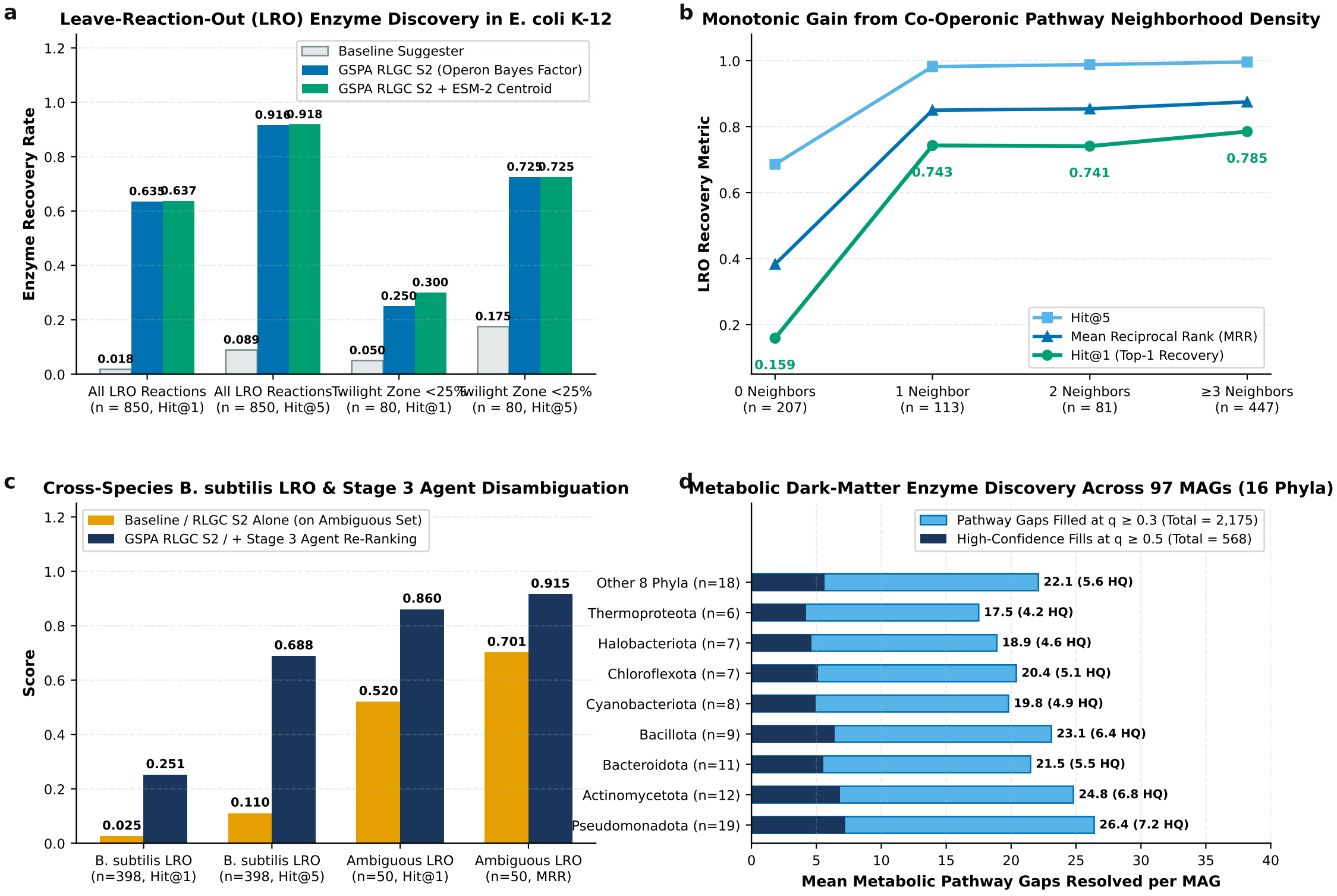
a
Leave-Reaction-Out (LRO) Enzyme Discovery in E. coli K-12
Baseline Suggester
GSPA RLGC S2 (Operon Bayes Factor)
GSPA RLGC S2 + ESM-2 Centroid
Enzyme Recovery Rate
0.018
0.635
0.637
0.089
0.916
0.918
0.050
0.250
0.300
0.175
0.725
0.725
All LRO Reactions (n = 850, Hit@1)
All LRO Reactions (n = 850, Hit@5)
Twilight Zone <25% (n = 80, Hit@1)
Twilight Zone <25% (n = 80, Hit@5)
b
Monotonic Gain from Co-Operonic Pathway Neighborhood Density
LRO Recovery Metric
Hit@5
Mean Reciprocal Rank (MRR)
Hit@1 (Top-1 Recovery)
0.159
0.743
0.741
0.785
0 Neighbors (n = 207)
1 Neighbor (n = 113)
2 Neighbors (n = 81)
≥3 Neighbors (n = 447)
c
Cross-Species B. subtilis LRO & Stage 3 Agent Disambiguation
Baseline / RLGC S2 Alone (on Ambiguous Set)
GSPA RLGC S2 / + Stage 3 Agent Re-Ranking
Score
0.025
0.251
0.110
0.688
0.520
0.860
0.701
0.915
B. subtilis LRO (n=398, Hit@1)
B. subtilis LRO (n=398, Hit@5)
Ambiguous LRO (n=50, Hit@1)
Ambiguous LRO (n=50, MRR)
d
Metabolic Dark-Matter Enzyme Discovery Across 97 MAGs (16 Phyla)
Pathway Gaps Filled at q ≥ 0.3 (Total = 2,175)
High-Confidence Fills at q ≥ 0.5 (Total = 568)
Other 8 Phyla (n=18)
22.1 (5.6 HQ)
Thermoproteota (n=6)
17.5 (4.2 HQ)
Halobacteriota (n=7)
18.9 (4.6 HQ)
Chloroflexota (n=7)
20.4 (5.1 HQ)
Cyanobacteriota (n=8)
19.8 (4.9 HQ)
Bacillota (n=9)
23.1 (6.4 HQ)
Bacteroidota (n=11)
21.5 (5.5 HQ)
Actinomycetota (n=12)
24.8 (6.8 HQ)
Pseudomonadota (n=19)
26.4 (7.2 HQ)
Mean Metabolic Pathway Gaps Resolved per MAG